

AI for Financial Practitioners

Lecture 0 · Part II

The Intelligence System: How Different Forms of Intelligence Work Together

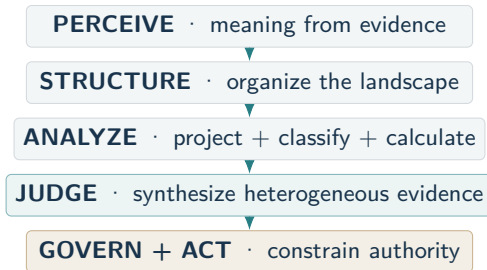
From the ladder of methods to a governed constellation of intelligence

Part I was the ladder — Part II is the constellation

INTELLIGENCE IS COMPOSITIONAL

Part I separated capabilities so we could understand them.

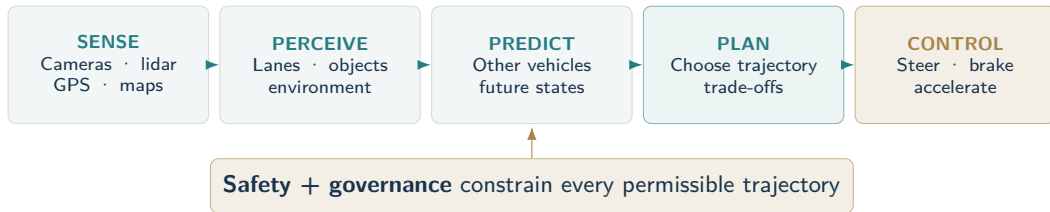
- Statistics, econometrics, ML, neural systems and generative AI remain useful.
- New methods expand the feasible problem set; they do not erase older ones.
- Real institutions rarely face a problem that belongs to only one analytical family.



Executive question: which capability should carry each responsibility, and how should the transitions be governed?

The autonomous-car analogy makes the architecture visible

NO SINGLE COMPONENT IS “THE INTELLIGENCE”



Core idea. A self-driving car is intelligent because **sensing, prediction, planning, control and safety constraints are coordinated**. Financial intelligence follows the same logic.

The AtlasPay mandate: one M&A problem, many forms of intelligence

A SYNTHETIC ACQUISITION UNIVERSE MAKES THE DIVISION OF COGNITIVE LABOR VISIBLE

AtlasPay seeks acquisition candidates across payments and software.

- 240 synthetic targets
- Merchant software · fraud/identity · cross-border · embedded finance
- Structured financial attributes + narrative dossiers
- Objective: identify candidates worthy of governed diligence

Narrative evidence · business model · product · geography

Operating evidence · revenue · growth · EBITDA margin

Quality / risk · recurring revenue · leverage · concentration

Market evidence · valuation multiple · adjacency

Institutional question · what should move to diligence?

The mandate requires semantic reading, market structure, forecasting, classification, exact calculation, judgment and authority.

The complete architecture: eight responsibilities, eight verbs

PERCEIVE → STRUCTURE → PROJECT → CLASSIFY → CALCULATE → JUDGE → GOVERN → ACT

01 · PERCEIVE

LLM extracts semantic observations from dossiers

02 · STRUCTURE

Clustering maps the acquisition landscape

03 · PROJECT

Regression estimates forward economics

04 · CLASSIFY

Neural network identifies nonlinear strategic fit

05 · CALCULATE

Deterministic finance anchors valuation + risk

06 · JUDGE

LLM synthesizes heterogeneous evidence

07 · GOVERN

Rules enforce evidence + authority boundaries

08 · ACT

Bounded workflow creates a diligence artifact

Perceive, structure, project: three different analytical claims

DO NOT BLUR MEANING, DISCOVERY AND ESTIMATION

PERCEIVE

Mechanism: Claude Haiku reads the target dossier.

Output: semantic attributes such as strategic relevance, product adjacency and concerns.

Claim: what does the narrative evidence mean?

STRUCTURE

Mechanism: unsupervised clustering.

Output: numerical peer groups and acquisition-landscape structure.

Claim: what groups exist without predefined labels?

PROJECT

Mechanism: linear regression.

Output: forward EBITDA-margin estimate with held-out error.

Claim: what operating economics are conditionally expected?

Transparency requires every output to retain its claim type, evidence source and model lineage.

Classify, calculate, judge: learned patterns need deterministic anchors

FLEXIBLE INTELLIGENCE SHOULD NOT ABSORB RESPONSIBILITIES BETTER HANDLED BY EXACT TOOLS

CLASSIFY

Neural fit model

Learns nonlinear strategic-fit patterns from financial and adjacency features.

Probabilistic pattern recognition



CALCULATE

Deterministic finance

Enterprise value · affordability · leverage · concentration · valuation flags.

Exact formulas + controlled inputs



JUDGE

Generative synthesis

Reconciles fit, forecast, valuation, risk and semantic evidence.

Contextual interpretation

Core idea. The LLM is strongest when it **synthesizes governed evidence**; it should not invent the arithmetic, the policy state or the approval.

Governance is part of intelligence — not a wrapper around it

JUDGMENT AND AUTHORITY ARE DIFFERENT THINGS

A candidate can look attractive and still be impermissible.

- Evidence can be incomplete.
- Leverage or concentration can breach policy.
- Valuation can exceed affordability.
- The system can be uncertain or internally inconsistent.
- Human transaction approval can remain mandatory.

A governable system must be able to abstain, reject or escalate.

Evidence gate · required fields present?

Risk gate · within leverage / concentration limits?

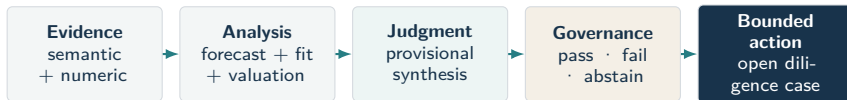
Affordability gate · financing capacity respected?

Confidence gate · enough support to proceed?

Authority gate · who may change institutional state?

Action should be bounded, reversible and auditable

AUTOMATE THE NEXT AUTHORIZED TRANSITION — NOT NECESSARILY THE FINAL DECISION



What the system may do

- Create a diligence initiation memorandum
- List analytical basis and unresolved risks
- Route approved workstreams to controlled workflow

What it does not do

- Commit capital or sign a transaction
- Invent approval status
- Convert uncertainty into action merely to finish the workflow

The executive playbook: design the intelligence system

FROM MODEL SELECTION TO INSTITUTIONAL ARCHITECTURE

01 · What must the system **perceive**, and from which evidence?

02 · What should be **estimated, classified or discovered**?

03 · What must remain **deterministic and exact**?

04 · Where does **contextual judgment** add value?

05 · What are the **governance, abstention and escalation** rules?

06 · What **bounded action** may the system legitimately take?

Objective: executable, transparent and governable intelligence.

The shift

From
“Which AI model should we use?”

To
“How should intelligence be assembled, governed and connected to action?”

Ladder → constellation → institution